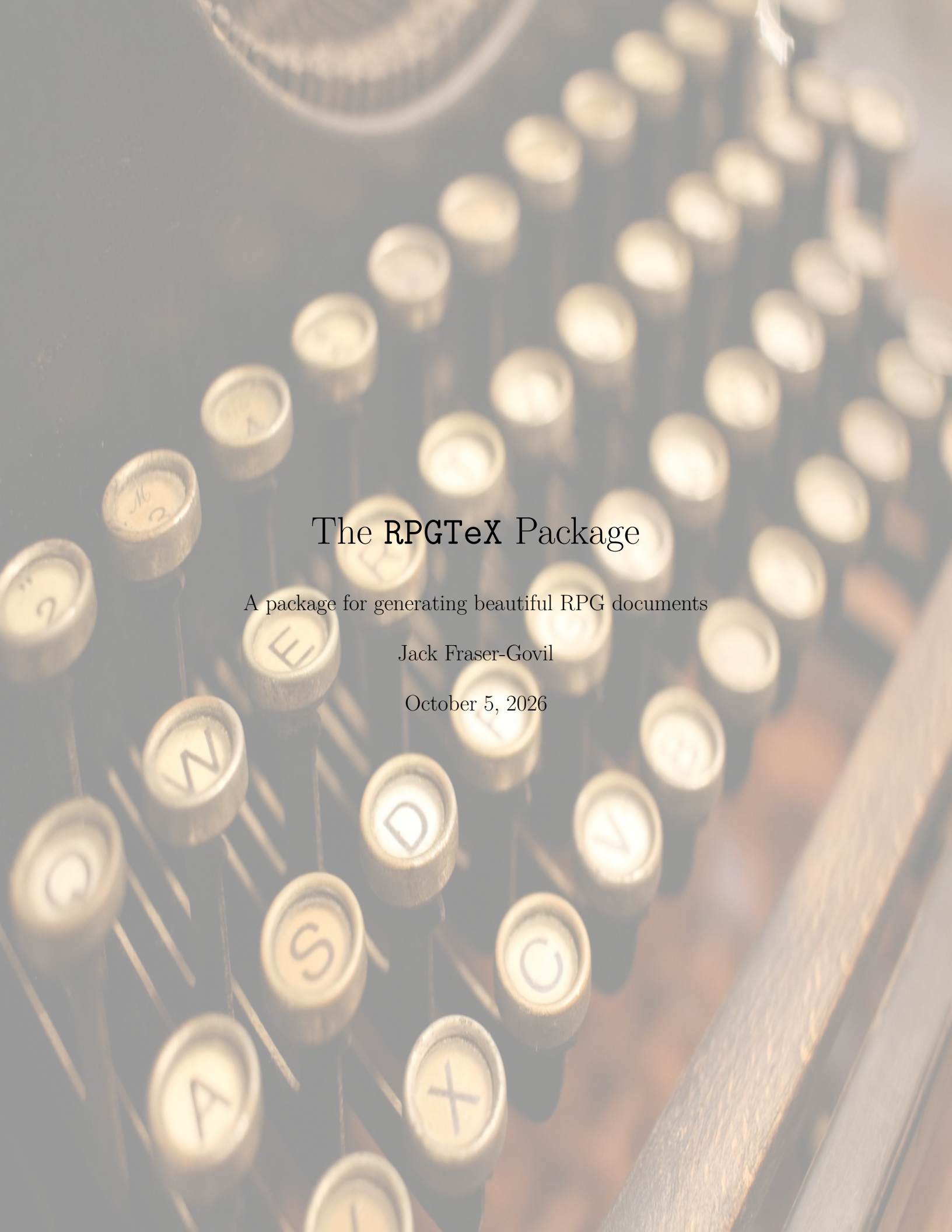




# The **RPGTeX** Package

A package for generating beautiful RPG documents

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**W**elcome to the *RPGTeX* package. This  $\text{\LaTeX}$  package is designed to allow users to flexibly typeset documents associated with Role Playing Games such as *Dungeons & Dragons* -- and many more besides. This package defines a central engine: `rpgcore` which define a number of useful functions and classes, and a flexible set of `themes` which control how those commands are rendered in the final document.

## Getting Started

This document is the full reference manual. For a quick example of how to get started with your first *RPGTeX* document, see the [example documents](#), which demonstrate the basic features of this package in a readable fashion.

## Attribution & License

This package would not have been possible without the team who developed [its predecessor, the 'DND 5e LaTeX Template'](#). That code was released under an MIT license, the text of which can be found in the `LICENSE` file. `rpgtex` is released under the same license.

## AI Usage Declaration

The author of this package used a handful of AI chat tools over the course of the development to help guide the thoughts of the human author. No automated coding tools were used anywhere in this package, and all code was written by hand. All mistakes and bad decisions are organic in nature.

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# Chapter 1: Installation & Usage

## Getting R<sub>P</sub>G<sub>T</sub>E<sub>X</sub>

There are a number of different ways to acquire R<sub>P</sub>G<sub>T</sub>E<sub>X</sub>. Once you have installed it, it is vital to ensure that it is properly configured (see below).

### Direct Installation

The simplest way to use R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> is to install it on the `texmf` path, where the compiler can automatically find it:

bash

```
mkdir -p "$(kpsewhich -var-value TEXMFHOME)/tex/latex" %make sure the directory exists
git clone https://github.com/DrFraserGovil/rpgtex.git "$(kpsewhich -var-value
TEXMFHOME)/tex/latex/rpgtex"
```

This will clone the repository into your L<sup>A</sup>T<sub>E</sub>X path.

### Indirect Installation

If you want to tinker with R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> -- such as by creating a new theme -- it is helpful to have it in a more accessible location. Clone the repository into a location of your choice:

bash

```
git clone https://github.com/DrFraserGovil/rpgtex.git /your/rpgtex/directory
```

You then have two options to make the package visible to the compiler:

### Use TEXINPUTS

Setting the environment variable `TEXINPUTS` allows the compiler access:

bash

```
TEXINPUTS=~ /your/rpgtex/directory/::
```

(Or similar commands, depending on your shell).

### Use Symlinks

You can symlink the install location to the `texmf` directory, allowing the compiler to act as if you had performed the `texmf` installation:

bash

```
mkdir -p "$(kpsewhich -var-value TEXMFHOME)/tex/latex"
ln -sf ~/your/rpgtex/directory "$(kpsewhich -var-value TEXMFHOME)/tex/latex/rpgtex"
```

### Overleaf (Not recommended!)

We do not recommend using Overleaf since the free-tier subscription has reduced compilation times drastically, making compiling documents using complex packages such as this one extremely difficult. Nevertheless:

1. Download this GitHub repository as a ZIP archive using the Clone or download link above.

2. On Overleaf, click the New Project button and select Upload Project. Upload the ZIP archive you downloaded from this repository.
3. Manually create the file `rpg-config.cfg` with the contents `\edef\RpgPackagePath{}`. This replaces the configuration step described below.

## Configuring RPGTeX

Because R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> comes packaged with some resources (namely some fonts) for which T<sub>E</sub>X does not support relative-pathfinding, it is necessary to provide R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> with a value for the `\RpgPackagePath`.

### `\RpgPackagePath` {} (Variable)

A variable holding the absolute path to the root directory of the rpgtex install location.

We provide a number of different ways to set this variable:

1. The provided `rpglatex` compiler automatically inserts a definition based on a `kpsewhich`, using the Command Line Interface (page ??).
2. If the variable does not exist, the R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> engine itself tries the following
  - (a) Searches for the file `core/rpg-config.cfg`, which (hopefully) contains a definition<sup>1</sup>, and `\inputs` it.
  - (b) If no file is found, R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> used the `\write18` utility to dispatch its own calls to `\kpsewhich` and build a config file in the expected location. This requires the compiler be run with `-shell-escape` active
  - (c) If this doesn't work, the compiler exits, and suggests that the user manually creates the config file:

```
cd <rpgtex root directory>
cmd="\edef\RpgPackagePath{$(pwd)}"
echo $cmd > core/rpg-config.cfg
```

3. We provide a shell script which executes these commands for the user.

The extra configuration steps need only be run once -- once the config file exists, the package will be able to configure itself.

### Why is configuration necessary?

T<sub>E</sub>X is generally set up so that when a file calls `include` or `input` it is possible to use filepaths relative to the package itself. `package.sty` can call `\input{dir/subpackage.tex}` and it will know to first check for the file relative to `package.sty`; even if the package resides within the texmf path and the user has no idea where `package.sty` is.

An annoying exception to this is fonts and typefaces. `xelatex` searches for fonts based on *filepaths relative to the current working directory* -- or from those installed in as system fonts.

Since R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> includes several (license free) typefaces as part of the provided themes, this poses a problem. We must either require that:

1. R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> documents can only be prepared in restricted locations relative to the install location of R<sub>P</sub>G<sub>T</sub>E<sub>X</sub>.
2. Users must identify and specify the R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> root path when preparing a document
3. Users must install the provided fonts to the system path
4. R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> must be configured to know 'where it is', and so provide an absolute filepath to the internal fonts.

The Configuration step is the most portable and easiest-to-use of these options.

## Package & Class Usage

R<sub>P</sub>G<sub>T</sub>E<sub>X</sub> can be used either as a standalone package, or as part of a number of classes

<sup>1</sup>In the form of `\edef\RpgPackagePath{path/to/package}`

## Standalone Package

The standalone package can be used directly by including the `RPGTeX` package:

```
\documentclass{arbitrary-class}
\usepackage[options]{rpgtex}
```

This will load only the core commands into the document, and (unless called explicitly) no themes will be imported. Using the package in this way does not activate any of the commands which change the overall geometry, background or headers of the document.

## Classes

RPGTeX can also be loaded through a number of classes which drastically alter the appearance of the document, defining new geometries backgrounds and adding headers.

The provided classes are:

1. `rpgbook` (page ??). Based on the standard book class, this is designed for larger RPG documents.
2. `rpghandout` (page ??). Based on the article class, this is designed for shorter documents
3. `rpgcard` (page ??). A small-document class designed for creating modular 'handout' cards for items, spells or abilities.

## Compiling rpgtex Documents

RPGTeX uses the `fontspec` package to allow custom fonts, and therefore requires compiling with `xelatex` or `lualatex`:

```
xelatex main.tex #works
lualatex main.tex #works
pdflatex main.tex #fails if fonts option passed
```

So long as RPGTeX is on the user's latex path, and the package properly configured (page 2) no further compilation steps are required. However, for ease of use, we provide the `rpglatex` compiler as part of the RPGTeX distribution.

## The rpglatex compiler

RPGTeX is shipped with a special compiler, `rpglatex`. This is simply a python3 script which acts as a wrapper around either `xelatex` or `luatex`, but includes several quality-of-life changes to the interface to make it easier to use with `rpgtex`.

```
rpglatex [options] <file>
```

`rpglatex` has the following features:

| Feature              | Description                                                                                                                                                                                                                                                                          | Options                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Compiler Selection   | The <code>xelatex</code> compiler is selected by default, but the <code>-l</code> , <code>-luatex</code> flags set it to use <code>luatex</code> instead.                                                                                                                            | <code>-l</code> , <code>-luatex</code> |
| Build Directory      | Compilation files ( <code>.aux</code> , <code>.log</code> etc.) are stored in a build directory. The default is <code>.build</code> in the calling location, but can be changed with the <code>-b</code> flag                                                                        | <code>-b &lt;build dir&gt;</code>      |
| Output Naming        | The name of the output file can be changed from the default (equal to the input tex name)                                                                                                                                                                                            | <code>-o &lt;output name&gt;</code>    |
| Multi-pass Compiling | By default, the compiler runs twice in a row to enable references and <code>tikz[remember]</code> commands to function. A full three-compilation suite (necessary for very complex or reference-heavy documents) can be activated with the <code>-f</code> , <code>-full</code> flag | <code>-f</code> , <code>-full</code>   |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Volume Control</b>     | <p>latex is notoriously noisy, producing copious output. By default, this is suppressed and only a summary is printed. The summary can be removed (rendering it completely silent) with the <code>-q</code> command, or the original output recovered in verbose mode; <code>-v</code>.</p> <p>These outputs are always overridden if a compilation error occurs, in which case the full trace is output to the console.</p> | <code>-q, -v</code>     |
| <b>Auto-bibtex</b>        | If the <code>-r</code> or <code>-ref</code> flag is set, <code>bibtex</code> is automatically called in between the multi-compilation steps                                                                                                                                                                                                                                                                                  | <code>-r, -ref</code>   |
| <b>Auto-visualisation</b> | If the <code>-show 1</code> option is set (which it is by default), the compiler will attempt to open the output file upon completion of the compilation; automatically opening or context-switching to the document. This can be turned off by calling <code>-show 0</code>                                                                                                                                                 | <code>-show</code>      |
| <b>Print Mode</b>         | A special interface for R <sub>P</sub> G <sub>T</sub> E <sub>X</sub> , this uses the <code>\RpgCMD</code> interface (page ??) to inject code into the latex document, setting the <code>bg=print</code> mode and suppressing the background output.                                                                                                                                                                          | <code>-p, -print</code> |

# PART I

## rpgtex For Users

# Chapter 2: Options & Variables

RPGTeX defines many dozens to hundreds of variables, most with the (expl3) syntax `\__rpg_[x]`. Most of these are used in the internal functioning of the macros, however a number of them are useful for a user to understand.

## Package Options

Whether the package is invoked directly, or through a class users have the option to pass options to it which change the behavior of the resulting document:

```
\documentclass[<options>]{rpgbook}
\usepackage[<options>]{rpgtex}
```

The options take one of two forms:

**Key-value arguments:** The name must be passed and a value set using an equals sign:

```
\usepackage[columns = 2, bg = print]{rpgtex}
```

**Flags:** Just the name has to be provided:

```
\usepackage[openany]{rpgtex}
```

The available option for the package are:

**background** <flag>

If present, activates the 'paper background' modules, including the background colour, paper images, geometry overloads and footer changes.

This flag is set by default when loading a class, but can be passed as a package option to enable the fonts.

**bg** = full / print / none

Controls the behaviour of the background and footer decorations. The value passed must be one of the three options (else an error is thrown). The most obvious effect of these three options is to change whether the 'background paper' set by `\RpgThemeSetPaper` (page ??) and `\RpgThemeBackgroundColor` (page ??) appears in the document, or the footer decorations set by `\RpgThemeFooterDecoration` (page ??).

| Command | Paper | Footer Decorations |
|---------|-------|--------------------|
| full    | ✓     | ✓                  |
| print   | ✗     | ✓                  |
| none    | ✗     | ✗                  |

This flag also changes the behavior of the `rpgcard` class (page ??), and other environments may similarly change their colours or layouts in response to the values passed to this command.

**columns** = 1 / 2

Sets the paper to either `onecolumn` or `twocolumn` mode. More advanced column-settings would require the user manually using `multicols`.

**font** <flag>

If present, activates the font modules, applying the package fonts wherever possible.

This flag is set by default when loading a class, but can be passed as a package option to enable the fonts.

**justified** <flag>

If present, sets the text to justified throughout the document. Default behaviour is left-aligned 'ragged right'.

**justified** <flag>

If present, sets the text to justified throughout the document. Default behaviour is left-aligned 'ragged right'.

**nomultitoc** <flag>

If present, disabled the multi-column table of contents option

**noemph** <flag>

Do not override the default behaviour of `\emph`, default behaviour aliases it to `\RpgKey`

**oneside** <flag>

Forces the document into oneside mode. Disables alternating footers but allows consistent asymmetric margins (such as in this document, where the left margin is larger to accommodate macro names)

**openright** <flag>

Places the document into right-opening mode (see book class documentation for more)

**sections** <flag>

If present, activates the sections modules, including the part-page overloads and any theme behaviour which modifies the behaviour of section commands.

This flag is set by default when loading a class, but can be passed as a package option to enable the fonts.

**size** = <font-size> pt

Sets the size equal to the value, if allowed by the parent class. The allowed values depend on the class being used (see the class section, beginning on page ??).

**theme** = <theme name>

The initial value passed to `\RpgSetTheme` (page ??) when package initialization is finished.

The default value is `default`, activating the Default Theme (page ??).

**themepath** = </path/to/theme/>

Calls `\RpgSetThemePath` (page ??) before the theme is loaded, allowing for custom themes to be set.

Default value is `\RpgPackagePath/themes`, with the assumption that the theme 'name' is stored in `themes/name/name.cfg`.

**title** <flag>

If present, activates the titling module and any theme-presets for altering the behaviour of `\maketitle`.

This flag is set by default when loading a class, but can be passed as a package option to enable the fonts.

## PART II

### rpgtex For Designers

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